

Data Science Challenge

George Boufis / f3352504

Publication Venue Prediction

Introduction:

The goal of the challenge is to predict the publication venue of a scientific paper among 7 possible classes. The available information consists of paper abstracts, publication years, author information, and a citation network connecting the papers. This is a challenging multiclass classification problem because useful information is distributed across multiple data sources. While the abstract contains semantic information about the research topic, the citation network provides structural relationships between papers, and the authors and publication year may also contain valuable predictive signals. To address this problem, several feature extraction and machine learning techniques were explored. The final approach combines textual representations, graph-based embeddings, metadata features, and ensemble learning methods in order to exploit all available information and improve classification performance.

The labeled training set contains 174,286 papers and the test set contains 43,650 papers. In the notebook was used a stratified 80/20 train/validation split, producing 139,428 training papers and 34,858 validation papers. Stratification is important because the class distribution is not uniform, preserving the label proportions in both splits makes the validation log loss a more reliable estimate of generalization performance. All experiments are evaluated using multi-class log loss.

Data Preparation:

The first step was to load the core files from the challenge dataset. The abstracts file uses a custom pipe-separated format where each line contains a paper id and abstract text separated by “||”, so the notebook defines a custom loader that splits each line into structured columns. Labels are loaded from y_train.txt, test ids from y_test.txt, and years from years.txt. After merging abstracts and years into the labeled and test data frames, the notebook checks missing values and finds no missing abstracts or years in either train or test. The final merged training table contains paper id, label, abstract, and year.

A fixed random seed is used throughout the notebook for Python, NumPy, and PyTorch. This improves reproducibility for the data split, neural-network initialization, mini-batch ordering, graph-walk generation. The notebook also detects CUDA and runs the neural models on GPU when available.

Leakage Prevention and Alignment:

The notebook handles every representation aligned with the same paper order by resetting indices after train/validation split and then deriving feature matrices from the corresponding id arrays.

The preprocessing steps also follow the standard rule of fitting transformations only on the training split. The author TF-IDF vectorizer and SVD model are fitted on training authors and then applied to validation and test authors. The year scaler is fitted on training years, and missing values are filled using the training median. The StandardScaler for the MLP is also fitted only on the fused training matrix.

The neighbor-label features require extra caution because they use labels from the nearby papers. The notebook avoids using validation or test labels by building the label lookup only from the training split. As a result, a validation paper can receive information from labeled training neighbors, but never from its own validation label or from other held-out labels.

Feature Representation:

The final representation is a 1,037-dimensional vector for each paper. It is built by horizontally stacking six feature groups: 768-dimensional SciBERT abstracts embeddings, 128-dimensional DeepWalk citation-graph embeddings, 5 graph-statistic features, 7 weighted neighbor-label features, 128 author features produced by TF-IDF followed by SVD, and 1 standardized publication year feature.

Text Features: SciBERT

For abstract text, the notebook uses precomputed SciBERT embeddings with dimensionality 768. SciBERT is a good fit because it was pretrained on scientific literature and therefore captures scientific terminology. A fine-tuned SciBERT model with LoRA adapters was explored and used to generate probability predictions and dense semantic embeddings. The LoRA setup targets the query and value matrices, uses low-rank adaptation, and evaluates accuracy, macro-F1, and log loss.

Citation Graph Features:

The citation graph is loaded from the edgelist.txt and represented as a directed NetworkX graph. The directed graph contains 217,936 nodes and 1,773,693 edges. Because citation direction matters, the directed graph is used for degree and PageRank-style features. An undirected version is also constructed for DeepWalk, where the goal is to learn neighborhood similarity rather than model citation direction explicitly.

DeepWalk is used to convert graph structure into dense embeddings. The generated walks are treated like sentences and trained with Word2Vec using vector size 128, window size 5, skip-gram training and 5 epochs. Finally, computes explicit graph statistics in-degree, out-degree, total degree, PageRank and clustering coefficient.

Author and Time Features:

Author information is loaded from authors.txt, parsed into paper ids and author strings, and cleaned by lowercasing, replacing commas and semicolons, removing unusual characters, and normalizing whitespace. The cleaned author strings are transformed with a word-level TF-IDF vectorizer using 1 to 3 grams. Because this representation is sparse and high-dimensional, truncated SVD reduces it to 128 dense dimensions.

Publication year is included as a single numerical feature. Missing years, if present after alignment, are replaced with the median year computed from the training split. The year feature is standardized

with StandardScaler, fitted only on the training data and then applied to validation and test. This avoids data leakage and prevents the raw year scale from dominating other features.

Neighbor-Label Features:

A particularly interesting engineered feature is the neighbor-label distribution. The notebook builds a lookup from training paper ids to known labels and then for each paper counts the venue labels of neighboring training papers in the citation graph. The counts are L1-normalized into a seven-dimensional probability-like vector. A weighted version is then used in the final feature matrix, outgoing cited neighbors receive weight 0.7, while incoming citing neighbors receive weight 1.0. The intuition is that papers connected by citations often belong to related communities, and incoming citations may be especially informative because they reflect later papers that considered this paper relevant.

Models and Training:

After feature fusion, the notebook trains several classifiers. The fused feature matrix is standardized before the MLP, producing training, validation and test matrices of shape 139,428 x 1,037, 34,858 x 1,037, and 43,650 x 1,037. The final system compares a neural fusion model, Logistic Regression, LightGBM, and precomputed SciBERT probabilities.

The Fusion MLP has two fully connected hidden layers with BatchNorm, ReLU activations, and Dropout, followed by a final linear layer for the seven class logits. The best search configuration is hidden sizes 512 and 256, dropout 0.4, learning rate 5e-4, and weight decay 1e-4. Then the best configuration according to the validation loss was retrained for up to 30 epochs with early stopping. A second MLP is also trained with a different random seed to reduce prediction variance when averaged in the final ensemble. Logistic Regression is used as a strong linear baseline on the same fused features. LightGBM is used as the main tree-based non-linear model. This gradient boosting algorithm is suitable here because it can capture non-linear interactions among text, graph, author and time features.

Validation Results:

Method	Validation Log Loss
Logistic Regression	0.4335
Fusion MLP best configuration	0.4053
Final Fusion MLP	0.4076
Second-seed MLP	0.4070
LightGBM	0.4038
Final ensemble	0.3815

The results show that the ensemble is substantially stronger than any individual model. LightGBM is the best single model, while the final weighted ensemble reaches 0.3815.

Experiments that did not improve Performance:

Several additional approaches were explored during the development process. Initially, TF-IDF representations of the abstracts combined with Logistic Regression were evaluated as baseline models. Although these methods provided reasonable results, they were outperformed by SciBERT embeddings. Other experiments included alternative ensemble configurations, CatBoost models, and 2-hop neighborhood features extracted from the citation graph.

Future Improvements:

A more extensive hyperparameter optimization process could be performed for both the neural network and boosting models. Furthermore, more advanced graph representation learning techniques, such as Graph Neural Networks (GNNs), could potentially capture richer structural information from the citation network. Finally, larger transformer models could be explored in order to further improve the classification performance.

Conclusion:

The project builds a complete classification pipeline for publication venue prediction using text, graph, author and temporal information. The final fused representation has 1,037 features per paper and supports several model families. Among individual models, LightGBM performs best with validation log loss 0.4038. The strongest result is the final ensemble, which averages two MLP runs and combines them with LightGBM and SciBERT probabilities, reaching validation log loss 0.3815. To conclude, venue prediction benefits from combining semantic content with citation-network and metadata signals, and that ensembling calibrated probability outputs gives a clear improvement over single-model predictions.